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MILITARY SPECIFICATION

PROJECTILES, 155-MM. GUN AND HOWITZER, M104 AND M110; WP, FS, HD, AND H FILLING ASSEMBLIES

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the white phosphorus (WP), chlorosulfonic acid—sulfur trioxide solution (FS), vacuum distillation process mustard (HD), and Levinstein process mustard (H) filling and assembly of the M104 and M110 155-millimeter projectiles without explosive components.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU–P–236 — Paper, Filter, Analytical.

UU–P–388 — Paper, Mimeograph.

MILITARY

JAN–P–215 — Phosphorus, White.

JAN–P–253 — Paper, Liquid-Vesicant-Detector, M6.

JAN–C–314 — Crayon, Vesicant (H or L) Detector, C14.

JAN–C–379 — Chlorosulfonic Acid—Sulfur Trioxide Solution (FS).

JAN–H–502 — Hexamine (For Use in Ammunition).

MIL–M–12051 — Mustard (HD) Vacuum Distillation Process.

MIL–M–12054 — Mustard, H (Levinstein Process).

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

FSC 1320

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DRAWINGS

ORDNANCE CORPS

75–14–296 — Shell, Chemical, M104, for 155-mm. Guns—Loading Assembly, Marking Diagram and Details.

75–14–317 — Shell, Chemical, M110, for 155-mm. Howitzer, M1—Loading Assembly, Marking Diagram and Details.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71–78 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–78 (Rev. 1950.")

3. REQUIREMENTS

3.1 **Materials and components.** All materials, components and subassemblies used in the filling assemblies shall conform to specifications and drawings forming a part of this specification.

3.2 **Filling.**

3.2.1 *WP.* When WP filling is specified (see 6.2), the projectiles, substantially clean and free from foreign matter, shall be filled with molten white phosphorus conforming to Specification JAN–P–215 (see 6.3). Whenever process water is used, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0 (see 3.3.1). The filling shall be solidified with the projectiles in a nose-up position.

3.2.2 *FS.* When FS filling is specified (see 6.2), the projectiles, dry and substantially free from foreign matter, shall be filled with chlorosulfonic acid—sulfur trioxide solution conforming to Specification JAN–C–379 (see 6.3 and 3.3.2).

3.2.3 *HD.* When HD filling is specified (see 6.2), the projectiles, dry and substantially free from foreign matter, shall be filled with distilled mustard conforming to Specification MIL–M–12051 (see 6.3 and 3.3.2).

3.2.4 *H.* When H filling is specified (see 6.2), the projectile, dry and substantially free from foreign matter, shall be filled with Levinstein mustard conforming to Specification MIL–M–12054 (see 6.3 and 3.3.2). Prior to closing the filled projectiles 48.0 ± 5.0 grams of hexamine conforming to Specification JAN–H–502 shall be added. No hexamine shall be permitted to remain on the press-fit surfaces in the nose of the projectiles.

3.3 **Filling height.** The projectiles shall be tested as specified in 4.4.2.

3.3.1 *WP.* The surface of the WP in the filled and closed projectile shall be no more than 1⅞ inches below the lowest point of the burster well adapter and the water above the WP shall be more than ⅜ inch below the lowest point of the burster well adapter when the projectile is inspected in a vertical nose-up position while the temperature of the WP filling is 130 ± 10°F.

3.3.2 *FS, HD, and H.* The distance below the bottom of the burster well adapter to the surface of the agent in the filled and closed projectile shall be within the limits of the

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applicable table when the projectile is inspected in a vertical nose-up position at the corresponding temperature range specified therein.

TABLE I. *FS—Filling level in closed projectile.*

Agent temperature (°F.)	Inspection level (inches)	
	Minimum	Maximum
40–49.9	3 3/16	4 1/16
50–59.9	3 1/16	3 15/16
60–69.9	2 15/16	3 7/8
70–79.9	2 13/16	3 13/16
80–89.9	2 11/16	3 11/16
90–99.9	2 9/16	3 9/16

TABLE II. *HD and H—Filling level in closed projectile.*

Agent temperature (°F.)	Inspection level (inches)	
	Minimum	Maximum
60–69.9	2 15/16	3 3/4
70–79.9	2 13/16	3 11/16
80–89.9	2 3/4	3 5/8
90–99.9	2 5/8	3 9/16

3.4 **Assembly.**

3.4.1 *M104.* Closing, coating, marking, and assembly of the M104 projectile shall be as specified on Drawing 75–14–296 except that the assembly shall not include the burster charge assembly, fuse well cup, and pad. Coating and marking for HD filled projectiles shall be the same as that for H filled projectiles except that the symbol "HD" shall be substituted for "H".

3.4.2 *M110.* Closing, coating, marking, and assembly of the M110 projectile shall be as specified on Drawing 75–14–317 except that the assembly shall not include the burster charge assembly, fuze well cup, and pad. Coating and marking for FS and HD filled projectiles shall be the same as that for WP and H filled projectiles respectively, except that the symbols "FS" shall be substituted for "WP" and "HD" substituted for "H".

3.5 **Cleaning and closing.** The exterior surface of the filled and closed projectile and the interior of the burster casing shall be thoroughly cleaned to remove any grease, oil, agent, or other contamination (see 6.5).

3.6 **Leakage.** The cleaned projectiles shall not leak when tested as specified in the applicable paragraph of 4.4.1.

3.7 **Pilot lot.** A pilot lot shall be produced in accordance with 4.1.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Lot.**

4.1.1 *Pilot.* Prior to the start of regular production and following any change in regular production methods, materials, or equipment, a pilot lot of 200 projectiles shall be filled using the same methods, materials, and equipment as will be employed during regular production.

4.1.2 *Regular production lot.* A regular production lot shall consist of the projectiles filled by the same filling installation under essentially the same conditions. Only projectile bodies with the same metal parts lot interfix shall be represented in any one lot of filled projectiles. The inspection lot for nondestructive inspection, within the above limitations, shall be designated by the inspector. A sublot shall consist of projectiles having the same zone designation (see 5.1).

4.2 Sampling for nondestructive inspection.

4.2.1 *Pilot lot.* Sampling shall be conducted in accordance with Standard MIL–STD–105, using inspection level III. Ten projectiles shall be inspected in accordance with the drawing and specification requirements. The remaining projectiles shall be inspected in accordance with the classification of defects and Standard MIL–STD–105, using inspection level III.

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4.2.2 *Regular production lot.* Sampling shall be conducted in accordance with the classification of defects (4.3.3) and Standard MIL–STD–105.

4.3 Inspection.

4.3.1 *Components.* The inspector shall ascertain that all materials, components, or subassemblies of the projectiles covered by separate specifications, standards, or drawings are in accordance with their respective specifications, standards, or drawings prior to final acceptance of the filled projectiles.

4.3.2 *Packing and marking.* The inspector shall inspect the packing and marking for compliance with Section 5.

4.3.3 *Classification of defects.* Inspection of the projectiles shall be conducted as follows:

4.3.3.1 *Shell, chemical, 155-mm. M104 and M110—loading assembly.* (Dwg. 75–14–296 or Dwg. 75–14–317)

Categories and defects	*Acceptance standards*
Critical	
A. Leakage	4.4.1
B. Agent contamination	
Major	***AQL 1.0 percent defective***
1. Depth to burster casing incorrect	
2. Threads damaged or contaminated	
3. Bourrelet diameter (max.) after coating incorrect	
4. Filling excessive (agent, WP or water)	4.4.2
5. Marking (identification) incorrect	
6. Zone marking incorrect	
Minor	***AQL 2.5 percent defective***
51. Rotating band damaged	
52. Marking (identification) missing or illegible	
53. Filling insufficient (WP)	4.4.2
54. Threads not greased	
55. Lifting plug missing, frozen or insecure (not seated)	
56. Coating missing, inadequate or incorrect	
Incidental	***AQL 10 percent defective***
101. Contamination (other than filling agent)	

4.4 Tests.

4.4.1 *Leakage.*

4.4.1.1 *WP.* Following the final operation and while the filling is still molten, the filled and closed projectile shall be placed in a nose-down position or rolled on its side for a minimum period of 1 minute. The emission of smoke shall constitute leakage.

4.4.1.2 *FS.* During or following the final cleaning operation the filled and closed projectile shall be placed in a nose-down position or rolled on its side for a minimum period of 1 minute. The emission of smoke shall constitute leakage.

4.4.1.3 *HD or H.* After the final cleaning operation, a disk shall be inserted into the fuze well so that the coated side of the paper is in contact with the burster well adapter. The disk shall be cut either from detector paper conforming to Specification JAN–P–253 or from filter paper conforming to type I, class 2 of Specification UU–P–236 or from mimeograph paper conforming to type I of Specification UU–P–388. The filter or mimeograph paper shall be coated with crayon conforming to Specification JAN–C–314. The projectile shall be maintained on its side or in a nose-down position at a minimum temperature of 70°F. for a minimum period of 16 hours and the disk examined for discoloration which shall be considered evidence of leakage.

4.4.2 *Filling height.* The projectile shall be radiographed (see 6.2).

4.5 Pilot lot rejection criteria. If any projectile fails to meet the applicable drawing or specification requirements, the pilot lot shall be rejected. The cause of failure shall be ascertained, corrective action instituted, and another pilot lot produced and subjected to the same examination and testing as for its initial pilot lot. No regular production lot shall be accepted prior to approval of a pilot lot.

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5. PREPARATION FOR DELIVERY

5.1 Packing, level A. Unless otherwise specified, the filled and closed projectiles shall be packed for shipment or storage in the same manner as the empty projectiles were received and the pack shall conform to the Code of Federal Regulations 49 CFR 71–78. Projectiles shall be segregated by zone for shipping purposes.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use.

6.1.1 *M104.* The M104 projectile is intended for use with 155-mm. guns.

6.1.2 *M110.* The M110 projectile is intended for use with 155-mm. howitzers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Type projectile and filling agent required.

(c) The contractor will not be required to furnish radiographic equipment.

6.3 Precautions. Because of the hazardous nature of filling munitions with chemical agents, filling installations are advised to take all necessary precautions to protect personnel from injury while filling projectiles in accordance with this specification.

6.4 Filling aid. The following information is furnished as an aid to procurement and not as a criterion for filling. Projectiles filled with agent within the limits of the applicable table for the corresponding temperature range have been found to comply with 3.3 provided the projectiles meet the internal dimensional requirements of the applicable drawings. The limits of each table are expressed in inches from the nose of the projectile to the upper surface of the agent before the burster casing is inserted into the projectile.

6.4.1 *WP.* The WP filling in the projectile will weigh 14.7 ± 0.7 pounds.

TABLE III. *WP—Filling level in open projectile.*

Agent temperature (°F.)	Filling level (inches)	
	Minimum	Maximum
120–129.9	6¼	6 15/16
130–139.9	6 3/16	6⅞
140–149.9	6⅛	6⅞
150–159.9	6 1/16	6 13/16

6.4.2 *FS.* The FS filling in the projectile will weigh 14.8 ± 0.6 pounds.

TABLE IV. *FS—Filling level in open projectile.*

Agent temperature (°F.)	Filling level (inches)	
	Minimum	Maximum
40–49.9	8⅜	8¾
50–59.9	8¼	8 11/16
60–69.9	8⅛	8⅝
70–79.9	8	8½
80–89.9	7 15/16	8 7/16
90–99.9	7 13/16	8 5/16

6.4.3 *HD and H.* The Hd filling in the projectile will weigh 9.8 ± 0.4 pounds. The H filling in the projectile will weigh 10.2 ± 0.6 pounds.

TABLE V. *HD and H—Filling level in open projectile.*

Agent temperature (°F.)	Filling level (inches)	
	Minimum	Maximum
60–69.9	8⅛	8½
70–79.9	8	8 7/16
80–89.9	7 15/16	8⅜
90–99.9	7⅞	8 5/16

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6.5 **Decontamination.**

6.5.1 *WP.* Projectiles may be decontaminated by passing through a hot water and detergent spray washer or by flushing with hot water while vigorously scrubbing with steel wool. Additional cleaning of the burster well may be effected with a steam lance and a power driven wire brush.

6.5.2 *FS, HD, or H.* Projectiles may be decontaminated by washing with a caustic solution followed by a hot water or steam rinse. Other approved decontamination solutions may be used.

6.6 **Safety classification.** Projectiles filled in accordance with this specification carry the following Ordnance safety classification:

TABLE VI. *Ordnance safety classification.*

Agent	Chemical ammunition group	Storage compatibility group	Explosive hazard class
WP	C	A	2
FS	B	A	11
HD	A	A	11
H	A	A	11

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Chemical Corps
Navy—Bureau of Ordnance

Other interest:
Army—OM
Navy—*MC*

Preparing activity:
Army—Chemical Corps

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